Discrete Event Simulation Course Syllabus

This syllabus defines a 12-week course in discrete event simulation delivered across 24 sessions and 48 teaching hours. It sets out learning outcomes, the weekly schedule, laboratory expectations, assessment weightings and academic conduct rules. Students learn to model queueing and service systems, build event calendars, validate models against observations and report findings with appropriate uncertainty language. All figures in this document are simulated planning values for instructional design purposes.

Supporting records are presented in Table 3.

Supporting records are presented in Figure 2.

1 Course Design and Learning Outcomes

The course is organized around a build-measure-explain cycle. Each week pairs a short conceptual lecture with a hands-on laboratory, so students encounter theory and implementation together. By the end, learners should be able to translate a process description into an event-driven model, justify input distributions, and distinguish simulated outputs from causal claims about real systems. Instructors review drafts within 5 working days to keep feedback timely.

Supporting records are presented in Table 2.

1.1 Weekly Topic Sequence

Topics progress from foundational mechanics to interpretation. Early weeks cover event scheduling, random number generation and entity flow; middle weeks address input modeling, warm-up periods and replication; later weeks focus on validation, sensitivity analysis and communicating results. The sequence assumes no prior simulation software experience but expects comfort with basic probability and programming constructs such as loops and records.

Supporting records are presented in Table 1.

1.1.1 Event Scheduling and Queue Mechanics

This unit examines the event calendar as the engine of any discrete event model. Students implement a priority queue of events, trace entity arrivals and departures, and observe how service discipline changes waiting behavior. Laboratory exercises require a trace log showing each event time and state change, so that instructors can verify understanding of simultaneity and tie-breaking rules rather than only final averages.

1.1.2 Random Variates and Input Modeling

Students generate variates from uniform, exponential and empirical distributions, then fit candidate distributions to provided arrival and service records. The unit stresses

documenting assumptions and testing fit rather than selecting a distribution by convenience. Exercises ask learners to compare fitted models against held-out observations and to explain, in plain language, why a good fit does not by itself prove the underlying process behaves that way.

1.2 Validation and Sensitivity Checks

Phase	Concept focus	Lab emphasis	Deliverable
Opening	Event mechanics	Priority queue	Trace log
Middle	Input modeling	Distribution fitting	Fit memo
Closing	Validation	Sensitivity checks	Validation log
Final	Communication	Caveat writing	Project report

Table 1: Weekly topic sequence with conceptual and laboratory emphasis

Validation activities compare model behavior with recorded system observations and structured expert review. Students perform face validation, trace-driven checks and simple sensitivity sweeps on key parameters. The unit emphasizes reporting ranges and caveats alongside point estimates, and it explicitly separates what the simulation observed from what may be inferred about the real system. Deliverables include a validation log with decisions and rejected alternatives.

Supporting records are presented in Figure 1.



Figure 1: Validation checks and the evidence each check produces

1.2.1 Trace-Driven Validation Exercises

Students replay recorded system traces through their models and compare simulated outputs against observed values. Exercises use a sample of 240 observations drawn from 4 sites across 2022 to 2025, giving a realistic basis for discrepancy analysis. Teams document each mismatch, classify its likely source, and decide whether to adjust logic, parameters, or

accept the deviation with a stated caveat in the validation log.

1.2.2 Sensitivity Sweeps and Reporting Ranges

Sensitivity sessions vary key inputs across plausible bounds and record how outputs respond, such as output values moving from 79.55 to 119.76 and costs from 51.71 to 77.84. Students summarize results as ranges with caveats rather than single point estimates, and they practice separating what the simulation observed from what may be inferred about the real system when presenting findings.

2 Assessment and Laboratory Schedule

Assessment combines laboratory submissions, a midterm modeling task and a final project with a written report and short presentation. Weights and due dates appear in the schedule table below. Late work is accepted within 2 days of the deadline with a documented penalty, and regrade requests must be filed within 5 days of feedback release. The final project replaces a traditional closed-book examination.

Supporting records are presented in Figure 3.

2.1 Assessment Components and Weightings

Each component targets a distinct skill: laboratories test implementation discipline, the midterm tests modeling judgment under time pressure, and the final project tests end-to-end reasoning from problem framing to reported conclusions. Rubrics are published at the start of term. Students receive formative feedback on at least two laboratory submissions before the midterm, ensuring they can adjust before higher-stakes work begins.

2.1.1 Laboratory Submissions

Laboratories are submitted as reproducible scripts plus a short memo. The memo must state model scope, input assumptions, replication count and observed output ranges, using the reporting template distributed in week one. Graders check that scripts run without manual edits and that memos avoid overclaiming. Partial credit is available for correct mechanics with incomplete interpretation, reflecting the course emphasis on both building and explaining.

2.1.2 Final Project and Presentation

The final project asks teams to model a chosen service process, validate it against supplied or collected observations, and present findings with explicit limitations. Presentations are brief and followed by questions on assumptions and alternative explanations. Teams must show at least one sensitivity check and one rejected modeling choice, demonstrating that conclusions were tested rather than asserted. A shared repository preserves scripts and logs for review.

2.2 Policies and Support Resources

The course follows institutional policies on academic integrity, accessibility and respectful conduct. Students may collaborate on conceptual discussion but must submit individually authored laboratory memos unless a project is explicitly team-based. Office hours run weekly, and a discussion forum supports asynchronous questions with responses typically within 2 days. Accommodation requests are handled through the standard institutional process and kept confidential.

Topic	Outcome	Lab focus	Evidence
Scheduling	Build models	Event trace	Trace log
Input modeling	Justify inputs	Distribution fit	Fit memo
Validation	Test models	Sensitivity sweep	Validation log
Reporting	Explain limits	Caveat writing	Final report

Table 2: Mapping of weekly topics to learning outcomes and laboratory work

Parameter	Planned value	Owner	Review point
Term length	12 weeks	Coordinator	Orientation
Sessions	24	Coordinator	Midterm
Teaching hours	48	Coordinator	Term end
Response window	2 days	Instructor	Weekly

Table 3: Course-level planning parameters used throughout the syllabus



Figure 2: Assessment components with weightings and feedback timing



Figure 3: Weekly assessment and laboratory schedule with due windows